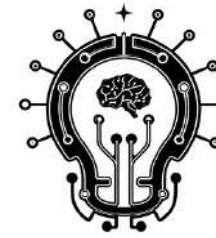


ABOUT THE PROGRAM

This program introduces students to humanoid robotics by exploring the structure and movement of the human hand. Students learn how to replicate these movements using mechanical design, servo motors, and basic electronics. The program combines engineering concepts with creativity, enabling students to design and build functional robotic prototypes.

EXPLORE TECHNOLOGY

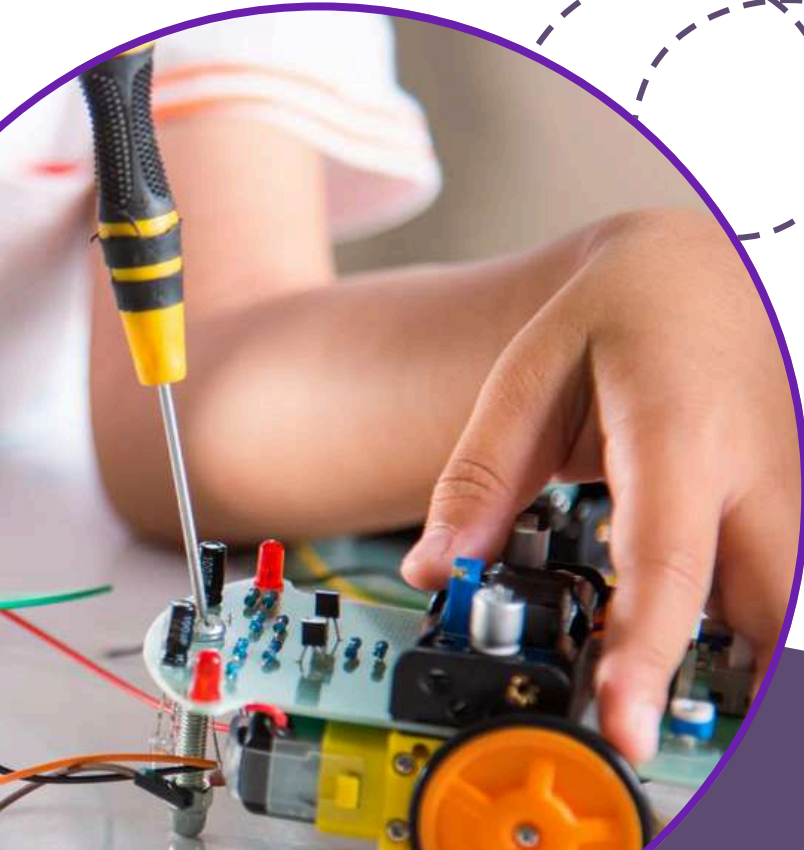
With Us



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GRADE 9
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IMPACT X
TECHNOLOGY &
INNOVATION



WHY THIS PROGRAM?



Introduces advanced robotics concepts



Connects biology with engineering



Focus on design and prototyping



Enhances analytical thinking



Encourages engineering creativity



Hands-on mechanical learning



Exposure to real-world applications

WHERE IDEAS TURN
Into Real Robots



PROGRAM OUTCOMES

- Understand human hand mechanics
- Design robotic voice and internet based pick and place bot
- Apply servo-based motion control
- Build functional robotic prototypes
- Present engineering-based solutions



PROGRAM FLOW



Advanced Foundations:

Electronics concepts, Arduino/ESP32 basics



Mechanics & Design:

Finger joints, linkages, and palm structure



Motion & Control:

Servo motors and wrist movement through voice and AI based



Electronics & Systems:

Basic circuits and microcontroller concepts



Prototype Development:

Design and build robotic wrist and fingers



Final Project:

Working prototype and presentation

FOR STUDENTS
OF GRADE 9